What is Controller Manager?

Professional Answer

The Controller Manager runs control loops that continuously monitor cluster state and ensure the actual state matches the desired state.

Unprofessional Answer

Controller manages Kubernetes.

Examples of Kubernetes controllers

Professional Answer

- Node Controller
- ReplicaSet Controller
- Deployment Controller
- Endpoint Controller

Unprofessional Answer

Deployment controller.

What does kube-controller-manager do internally?

kube-controller-manager runs multiple controllers that manage different cluster resources.
Each controller continuously compares the desired state with the actual state.
If a mismatch is detected, corrective actions are taken automatically.
Examples include ReplicaSet, Node, Deployment, and Endpoint controllers.
This reconciliation mechanism enables Kubernetes self-healing behavior.

kube-controller-manager (Reconciliation Engine)

Internal Design

- Runs **multiple controllers**
- Each controller is a **control loop**

Control Loop Logic

1. Watch desired state (API server)
2. Compare with actual state
3. Take corrective action
4. Repeat forever

Key Controllers Internals

- **ReplicaSet Controller**
  - Creates/deletes pods
- **Node Controller**
  - Detects node failure
- **Deployment Controller**
  - Manages rolling updates
- **Endpoint Controller**
  - Updates service endpoints

Core Principle

Kubernetes is a **reconciliation-based system**

What is kube-controller-manager?

kube-controller-manager is a **control-plane component** that runs multiple **controllers**.
Controllers continuously watch cluster state and **reconcile actual state to desired state**.
It operates on the **control loop pattern**.
It does not run workloads.
It ensures Kubernetes is self-healing.

Why kube-controller-manager exists

Kubernetes is **declarative**, not imperative.
Users define **desired state**.
Controllers enforce that state continuously.
Without controllers, Kubernetes would not auto-heal.
This component is the **brain of reconciliation**.

How kube-controller-manager works

Each controller follows this loop:

1. Watch API Server
2. Compare desired vs current state
3. Take corrective action
4. Update API Server

What happens if kube-controller-manager goes down?

Feature	Impact
Existing Pods	Continue running
Auto-healing	Stops
Scaling	Stops
Node eviction	Stops
Jobs	Fail silently

Cluster becomes **non self-healing**.

4. Important controllers inside kube-controller-manager

1. Node Controller

- Monitors node health
- Marks nodes NotReady
- Evicts Pods when node is unhealthy

---

2. Replication Controller

- Ensures correct Pod count
- Recreates deleted Pods
- Maintains availability

3. Deployment Controller

- Manages ReplicaSets
- Handles rolling updates
- Ensures deployment strategy

---

4. StatefulSet Controller

- Maintains Pod identity
- Handles ordered deployment
- Manages stable storage

5. DaemonSet Controller

- Ensures one Pod per node
- Handles node join/leave
- Used for agents (logging, monitoring)

---

6. Job & CronJob Controller

- Manages batch workloads
- Handles retries and failures
- Schedules periodic jobs

7. Endpoint Controller

- Updates Service endpoints
- Tracks healthy Pods
- Keeps traffic routing accurate

---

8. Namespace Controller

- Manages namespace lifecycle
- Handles deletion & cleanup
- Removes dependent resources

---

9. ServiceAccount & Token Controller

- Creates default service accounts
- Manages token secrets
- Enables Pod authentication